

ECON-1020H Introductory Macroeconomics

Chapter 6 Measuring the Cost of Living

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The Consumer Price Index

Using the CPI

Problems with the CPI

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This Week:

- Textbook Chapters 6 and 9

Next week:

- Textbook Chapter 7

The Consumer Price Index

Price indices and the aggregate price level

Two measures of aggregate price level:

- The GDP deflator = nominal GDP / real GDP
- The **Consumer Price Index** (CPI) measures the overall cost of goods and services bought by a typical consumer

Increases in price level are called **inflation**

Declines in price level are called **deflation**

When increases of the price level slow down, we call it **disinflation**

Inflation rate is the percentage change of the price level

CPI is measured in three steps:

- **Step 1.** Fix the basket of goods and services, assign weights to them; Find the price of each good and service in the basket at each point in time
- **Step 2.** Compute the price (the cost) of the basket, using the weights and individual prices of goods and services
- **Step 3.** Choose a base year and calculate the index as (see next page)

CPI in current year:

$$\text{CPI} = \frac{\text{Cost of basket of goods and services in current year}}{\text{Cost of basket of goods and services in base year}} \times 100$$

- The basket and weights must remain the same between two years

The **inflation rate** of CPI in year t is

$$\pi_t = \left(\frac{\text{CPI}_t}{\text{CPI}_{t-1}} - 1 \right) \times 100\%$$

Note: Time period does not have to be year, it can be month or quarter

The basket: prices and weights

In Canada, the CPI basket consists of eight aggregated items (categories)

- Shelter, food, clothes, transportation, etc.

Basket weights are based on relative expenditures across different items

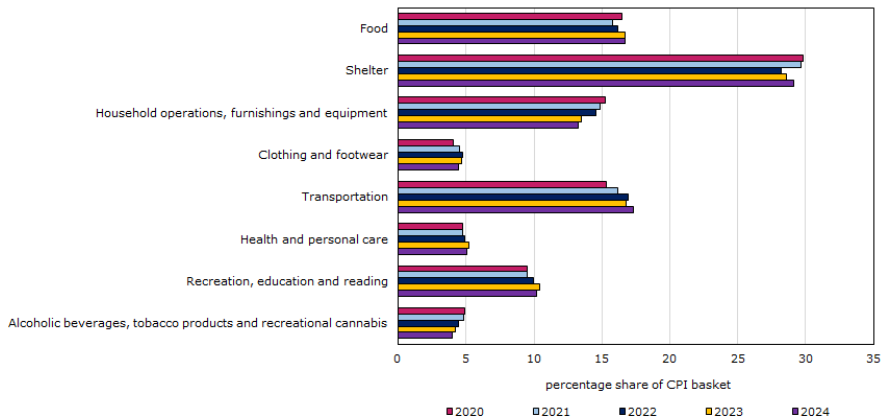
- Expenditure data:
 - [Household Final Consumption Expenditure](#)
 - Survey of Household Spending

Basket weights are normally updated annually. See Table A1 in [this article](#)

The **Monthly Price Survey** is the primary source of prices of goods and service in basket

The CPI basket, Canada 2020 to 2024

Chart 1
Basket share by major component, Canada, 2020 to 2024



Note: Weights are expressed as proportions of total household consumption expenditures in scope of the CPI in the weight reference period dollars.

Source: Statistics Canada, Table 18-10-0007-01 - Basket Weights of the Consumer Price Index.

Some Resources on CPI

[CPI Frequently Asked Questions](#)

[Personal Inflation Calculator](#)

[Understanding the Consumer Price Index](#)

An illustrative simple example

See next page for data

CPI is $\frac{70}{50} \times 100 = 140$ in 2018, and is 100 in 2015

Inflation rate from 2015 to 2018 is $\left(\frac{140}{100} - 1\right) \times 100 = 40\%$

See your textbook for another example

A Simplified Calculation

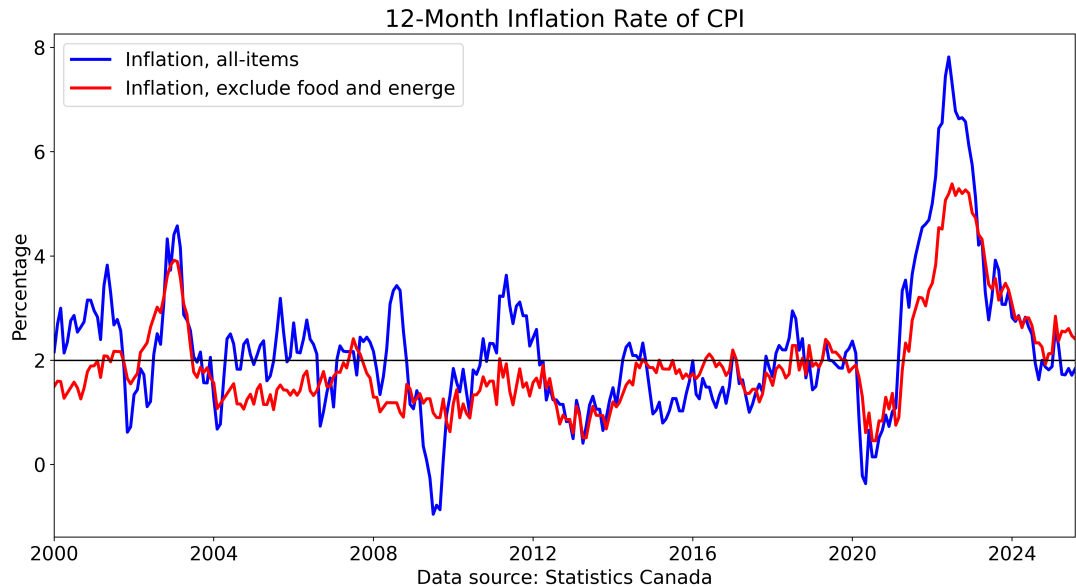
(a) The cost of the CPI basket at base-period prices: 2015

CPI basket		Price	Cost of CPI Basket
Item	Quantity		
Oranges	10	\$1	\$10
Haircuts	5	\$8	<u>\$40</u>
Cost of CPI basket at base-period prices			<u><u>\$50</u></u>

(b) The cost of the CPI basket at current-period prices: 2018

CPI basket		Price	Cost of CPI Basket
Item	Quantity		
Oranges	10	\$2	\$20
Haircuts	5	\$10	<u>\$50</u>
Cost of CPI basket at current-period prices			<u>\$70</u>

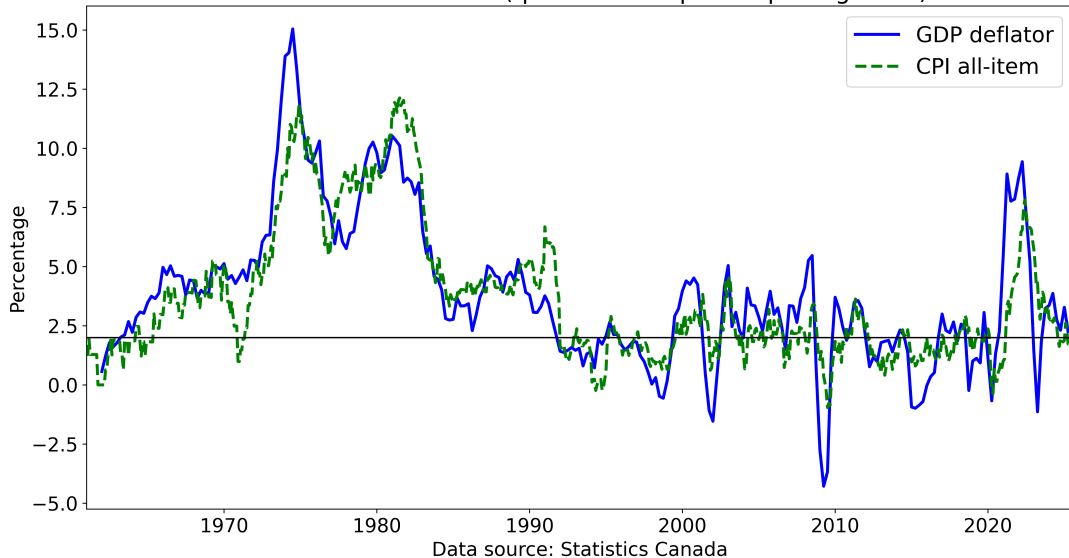
CPI inflation in Canada



Differences between CPI and GDP deflator

- CPI measures the price level of consumption goods
- GDP deflator measures the *implicit* price level of GDP
- The two price indices can diverge, not surprisingly
- Changes in prices of investment goods are not reflected in changes in CPI
- CPI does not reflect the relative export-import price
- Weights in the CPI basket is fixed, while the weights of goods in GDP are not fixed in real data

Annual inflation Rates (quarter-over-quarter price growth)



Using the CPI

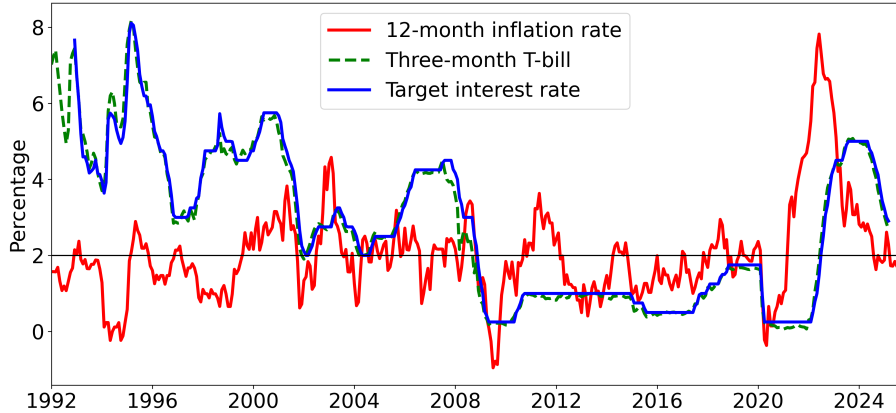
Getting real with the CPI

- Indexation: Correct a dollar amount for the effects of inflation by law or contract
- Calculating real values, e.g., real interest rate

$$\text{Real interest rate} = \text{Nominal interest rate} - \text{inflation rate}$$

- The equation above is an identity, it does not tell a causal relationship

Inflation and nominal interest rates



Data source: Statistics Canada

Problems with the CPI

Problem with CPI

CPI is an imperfect measure of changes in the cost of living

Three major problems leading to measurement biases:

- Substitution bias
- Introduction of new goods
- Quality improvement poorly or under-measured

These biases have been more severe since around 2003

Substitution bias

Prices of goods and services in the CPI basket do not change proportionately

- Some prices rise more than others

Consumers substitute towards those goods whose prices become relatively lower

They purchase more of goods with smaller price increases, and less of goods with larger price increases

But, in the CPI basket, the weights of goods and services are fixed, ignoring substitution

Thus, CPI over-measures increases in the cost of living, an upward bias

The weights of goods with a larger price increase are fixed in CPI basket, but may become smaller in the economy

Bias due to new goods

When a new good is introduced, consumers have more variety to choose from. This reduces the cost of maintaining the same level of economic well-being, because each dollar is worth more.

Why is that? Consider the introduction of smart phones. A smart phone reduces the cost of living in the sense that a new variety allows consumers to choose more and better for the same income level.

But CPI basket does not include new goods, CPI measure is biased upward.

Bias due to under-measured quality change

If the quality of a good rises over time, the value of a dollar is higher, because consumers get more of a good for the same amount of money

- E.g., adding a cruise control to a car is an increase in quality, which may not lead to an increase in the price of cars, but a cruise control makes the cost of living smaller

Statistical agencies often adjust prices to account for the quality change

Despite the effort, quality improvement remains hard to measure, likely causing another upward bias of CPI

Bias may have become more severe since mid-2000s

According to research by economists, mis-measurement of CPI and inflation was accelerated starting around 2003

Quality, not quantity, has become more important in output growth (through product expansion and turnover)

- E.g., self-driving cars, streaming entertainment, artificial intelligence

Big 4 corporations (Apple, Google, Amazon, Facebook) introduced new products since 2005

- iphone (2007), ipad (2010), Google maps (2005), Youtube (2005), AWS (2006), Alexa (2012), ...

Many of these new consumer products do not show up in CPI and inflation, because they are "free"

They also improve quality by replacing old products, e.g., CDs and cameras, thus they are "output saving"

Intangible investment replaces tangible investment

- E.g., big data, machine learning algorithms, genome sequencing, cloud computing, SpaceX

The cost of these investment declines fast

Some estimate that these rapid and new developments may have led to a zero inflation in the U.S

- while measured inflation has been close to 2% lately (before 2020) in the U.S.